natural phenomena—rest and motion, earth and fire—but Boyle associated this notion with chemical processes in a thoroughly materialistic way. Chemistry was able to convert different substances into each other. Boyle's query was: From what substances can the infinite variety of homogeneous substances existing in nature be built up? And furthermore: What are the elements that cannot be resolved any further, and of which all substances are composed, in one way or another? This problem arose out of the originally quite different question raised by the alchemists in the centuries before Boyle. Alchemy had developed out of the fundamental idea that every substance could ultimately be reduced to one basic substance, and that it must be possible, in principle at least, to convert any substance, any type of matter, into any other—mercury into gold, for instance. But all efforts in this direction had always remained futile; such transmutation could never be effected by chemical means. It appeared obvious that matter was not homogeneous in this sense—when treated by chemical means—but there had to exist basic substances which no chemical process could change into another. Since Boyle's time it has become a matter of common knowledge that there exists a whole series of these basic substances, or *chemical elements,* as against the approximately half a million uniform *chemical compounds* known to-day. The number of chemical compounds exceeds by far that of the basic elements. Nevertheless, the number of the elements is still large enough to make it difficult for us to regard them as the ultimate, indivisible building blocks of matter. Of course, Boyle knew relatively only few of the ninety-six elements known to us to-day, but nevertheless he succeeded in formulating quite clearly the aims and tasks of chemistry. He said: 'What we have to do is to determine into what basic substances matter can be analysed by chemical means, and what these basic substances are.' Thus we see that his chemical elements had nothing more in common with earth, air, fire and water, the elements of Democritus.

A century later came Lavoisier, the real father of modern chemistry. He was born in 1743 and died, a victim of the French Revolution, in 1794. His permanent contribution to science was the founding of quantitative chemistry. He was first to interpret rightly the process of combustion. Up to his time, it had been believed that in the combustion of any body a substance called *phlogiston* was released, and therefore, bodies would necessarily become lighter after combustion. Lavoisier adopted the opposite view, that combustion consisted in the combination of an element with oxygen, and as a result, the body must become heavier. His theory was proved correct by experiment. At the same time, he accomplished something of vast importance, in that he stimulated chemists to investigate changes in mass due to chemical changes.

Now we come to a law which was formulated by Lavoisier in 1774, but